Rachel Gross – Curriculum Vitae
University of Virginia, Department of Astronomy
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Research Interests

Gas- and ice-phase chemistry in star- and planet-forming environments, with a focus on how irradiation, temperature, and physical structure influence molecular abundances and observable signatures. My research combines laboratory experiments, astronomical observations, chemical modeling, and radiative-transfer simulations to investigate the physical and chemical processes shaping these environments.

Academic Appointments

ICI Postdoctoral Fellow, Department of Astronomy, University of Virginia 2026–Present

Education

Ph.D. in Chemistry, University of Virginia May 2026

Dissertation: "UV- and Thermally-Driven Chemistry in Star and Planet Formation: An Experimental, Observational, and Theoretical Approach"

B.Sc. in Chemistry, Minor in Mathematics, University of Houston-Downtown December 2019

Summa cum laude

Awards and Fellowships

Interdisciplinary Doctoral Fellowship, University of Virginia 2020–2025

First Year Merit Award, UVA Department of Chemistry 2020

Welch Foundation Grant, Scholar's Academy & Nuclear Regulatory Commission 2019

Publications

- **Gross, R.E.**, Terwisscha van Scheltinga, J., Cleeves, L.I., Dukes, C.A., Garrod, R.T., *et al.*, 2026, *accepted*; arXiv:2609.27031 – "Laboratory Evidence that Methanol-Rich Ice Mantles Lower Methyl Formate Binding Energies."
- Thompson, W.E., Bergner, J.B., Evans, N.J., *et al.*, including **Gross, R.E.**, 2026, ApJ – "CORINOS V: Radiative Transfer Effects in Protostellar Ice Observations."
- Turner, A.M., Yang, Y.-L., **Gross, R.E.**, Sakai, N., Kaiser, R.I., 2026, ApJ – "A Chemistry-first Centered Icy Chemical Inventory of Protostellar Sources with JWST."
- **Gross, R.E.**, Yang, Y.-L., Cleeves, L.I., van Dishoeck, E.F., Garrod, R.T., *et al.*, 2026, ApJ – "CORINOS IV: Quantifying Baseline-fitting Uncertainties in SO_2 Ice Measurements with JWST/MIRI."
- **Gross, R.E.**, Cleeves, L.I., 2025, ApJ, 980, 189 – "Modeling the Impact of Moderate External UV Irradiation on Disk Chemistry."
- Williams, B.J., Cleeves, L.I., **Gross, R.E.**, Baker, J., 2025, ApJ, *accepted* – "Investigating the Sulfur Mystery in Protoplanetary Disks Through Chemical Modeling."
- Ballering, N.P., Cleeves, L.I., Boyden, R.D., McCaughrean, M.J., **Gross, R.E.**, Pearson, S.G., 2025, ApJ, 979, 110 – "Water Ice in the Edge-on Orion Silhouette Disk 114–426 from JWST NIRCam Images."

Awarded Observational Time

Principal Investigator

ALMA Cycle 12, 2025.1.00765.S, 9.3 hr
"Uncovering a Hidden Complex Nitrile Reservoir in Planet-Forming Disks."

Co-Investigator

JWST GO 10076, Cycle 5, 13.0 hr; PI N. Ballering
"I see a little silhouetto of a disk: Measuring protoplanetary ice and gas in absorption against a background nebula."

ALMA Cycle 11, 2024.1.00494.S; PI R. Boyden
"The Band 1 Orion Nebula Cluster Disk Survey."

JWST GO 6130, Cycle 3, 17.7 hr; PI R. Boyden
"The Most Typical Planet Formation in the Most Typical Environments."

Research Experience

Postdoctoral Research

Interconnected Cosmos Initiative (ICI), University of Virginia 2026–Present
Advisors: Prof. Ilse Cleeves (Astronomy) and Prof. Catherine Dukes (Laboratory for Astrophysics and Surface Physics)

- Coordinating the relocation and recommissioning of the UHV laboratory setup astrochemistry laboratory in a new facility
- Developing observational approaches to investigate ice survival in externally irradiated protoplanetary disks and studies of Orion silhouette disks
- Developing laboratory experiments to investigate how CO_2 ice porosity influences its infrared spectral properties, with applications to interpreting CO_2 observations and abundance measurements on Ganymede

Doctoral Research

Laboratory Astrochemistry and Instrumentation
Laboratory for Astrophysics and Surface Physics, University of Virginia. PI: Catherine Dukes

- Led a major overhaul of an existing cryogenic ultra-high-vacuum chamber ($\sim 10^{-10}$ Torr), upgrading cryogenic controls, electronics, pressure monitoring, LabVIEW software, and transmission FTIR capabilities
- Developed a custom quartz crystal microbalance (QCM) analysis pipeline to calibrate ice deposition rates and characterize sub- and multilayer deposition
- Independently maintained and operated the chamber and trained undergraduate researchers in laboratory instrumentation and experimental procedures
- Measured the desorption behavior and binding energies of complex organic molecules, including methyl formate on H_2O- and CH_3OH-rich substrates, using temperature-programmed desorption and QMS

JWST Ice Spectroscopy and Methodology
CORINOS Collaboration, PI: Yao-Lun Yang

- Analyzed JWST/MIRI spectra of four Class 0 protostars, focusing on weak and blended ice absorption features in the 7–8 μm region
- Developed a Monte Carlo baseline-fitting methodology using more than 50 randomized continuum fits per source to quantify systematic uncertainties in weak SO_2 ice measurements
- Combined continuum sampling with laboratory ice spectra to constrain ice absorption and derived molecular abundances

Protoplanetary Disk Chemistry and Modeling
Department of Astronomy, University of Virginia Advisor: Ilse Cleeves

- Extensive chemical models to investigate how moderate external UV irradiation and disk structure influence gas- and ice-phase molecular abundances

- Investigated molecular tracers, binding-energy prescriptions, and their effects on disk chemistry and snowline locations
- Coupled chemical and radiative-transfer models to generate synthetic observations and investigate how disk geometry, scattering, and spectral extraction influence observable molecular signatures

Previous Research Appointments

Institute of Applied Radiation Chemistry, Lodz, Poland Summer 2019

Research with Dr. Piotr Ulanski investigating radiation-induced crosslinking and synthesis of polymeric nanogels for wound-dressing applications. Gained experience with electron-beam accelerator procedures and radiation chemistry

Laboratory of Organic Electronics, Lodz, Poland Summer 2019

Research involving nanostructured TiO_2 thin-film fabrication for solar-cell applications. Techniques included atomic force microscopy, spin coating, glovebox operation, cleanroom procedures, and vacuum evaporation

Undergraduate Research, University of Houston-Downtown 2018–2019

Completed research projects in computational and inorganic chemistry, including:

- Density functional theory calculations involving lead-iodide perovskite crystals
- Computational investigation of van der Waals interactions and actinide systems
- Phthalocyanine metalation synthesis and characterization

Technical Experience

Laboratory Methods: Ultra-high vacuum, cryogenic ice deposition, temperature-programmed desorption, FTIR spectroscopy, quadrupole mass spectrometry, quartz crystal microbalance measurements, UV irradiation, and experimental calibration

Instrumentation: Vacuum hardware, cryogenic and temperature control systems, pressure monitoring, optical alignment, electronics troubleshooting, LabVIEW, and CAD

Computational Methods: Python, Fortran, chemical modeling, radiative-transfer modeling (including RADMC-3D), JWST spectral analysis, and statistical baseline-fitting methods

Research Mentoring

Undergraduate Research Mentor, University of Virginia 2023–Present

Mentored four undergraduate researchers across laboratory astrochemistry, instrumentation, and theoretical chemical modeling

- **Hanan al Abbar** – Experimental measurements investigating binding energies on mixed ice substrates
- **Shane Sawyer** – CAD modeling and instrumentation design for the UHV laboratory setup
- **Kyle Gresko** – UHV system operation and laboratory ice experiments
- **Becky Williams** – Protoplanetary disk chemical modeling using updated sulfur reaction networks

Teaching Experience

Teaching Assistant, General Chemistry Laboratory, University of Virginia 2020–2021

Undergraduate laboratory instruction, grading, and student support with office hours

Academic Service

Journal Club Co-Leader, Department of Astronomy, University of Virginia 2023–2024

Organized and moderated weekly research discussions on current literature in astrochemistry, planet formation, and observational astronomy

Science Communication and Outreach

Laboratory Tours and Demonstrations, University of Virginia 2025–2026

Coordinated and led tours of the astrochemistry laboratory for students in Ilse Cleeves's course and visiting Starhill students. Introduced participants to laboratory astrochemistry, experimental instrumentation, and current research activities

Go Explore the Universe, University of Virginia 2026

Summer astronomy outreach program. Co-developed and prepared a hands-on Radio JOVE activity introducing students to radio astronomy

Girls Exploring the Universe, University of Virginia 2022, 2024, 2025

Summer astronomy camp for middle school girls, led by UVA women astronomers. Contributed to the development and preparation of the Radio JOVE activity for the 2025 program

Astrochem Coffee Podcast 2025

Featured in the *Single-Origin Brew* segment discussing the impact of external UV irradiation on protoplanetary disk chemistry

Listen to episode

Dark Skies, Bright Kids, Charlottesville, VA 2021–Present

Hands-on astronomy education and outreach for elementary and middle school students

Conferences and Presentations

Invited Talks

- Center for Astrophysics, Institute for Theory and Computation Lunch Series, Cambridge, MA, April 2025 – "JWST MIRI Reveals SO_2 and CH_4 in Planetary Nurseries"
- Exoplanet Coffee Colloquia, University of Wisconsin–Madison, October 2024 – "Ice Desorption Experiments with Complex Organics at UVA"

Contributed Talks and Seminars

- DISCO: DISks in COntext Conference, NRAO, Charlottesville, VA, September 2026 – "Radiative-transfer modeling of externally irradiated disk ices"
- 245th American Astronomical Society Meeting, National Harbor, MD, January 2025 – "Mid-Infrared Ice Spectroscopy of Class 0 Protostars: Decoding SO_2 and CH_4 Contributions"
- 4th Year Seminar, UVA Department of Chemistry, March 2024 – "Photochemistry during Planet Formation: Impacts on the Gas and Ice"

Poster Presentations

- Origins of Solar System GRC, South Hadley, MA, June 2025 – "Blended Lines, Clear Insights: Quantifying Weak Ice Features in Protostars with JWST"
- New Heights in Planet Formation, ESO, Germany, July 2024 – "Photochemistry and Planet Formation: A Theoretical Study of Irradiated Protoplanetary Disks"
- Laboratory Astrophysics Workshop (ICE), Kauai, February 2024 – "Investigation of COM Desorption from Surfaces and Mixtures of H_2O, CO, and CH_3OH"
- Origins of Solar System GRC, South Hadley, MA, June 2023 – "Living on the Edge: Disk Chemistry in the Outskirts of Dense Star Forming Regions"
- Protostars and Planets VII, Kyoto, Japan, April 2023 – "Living on the Edge: Disk Chemistry in the Outskirts of Dense Star Forming Regions"